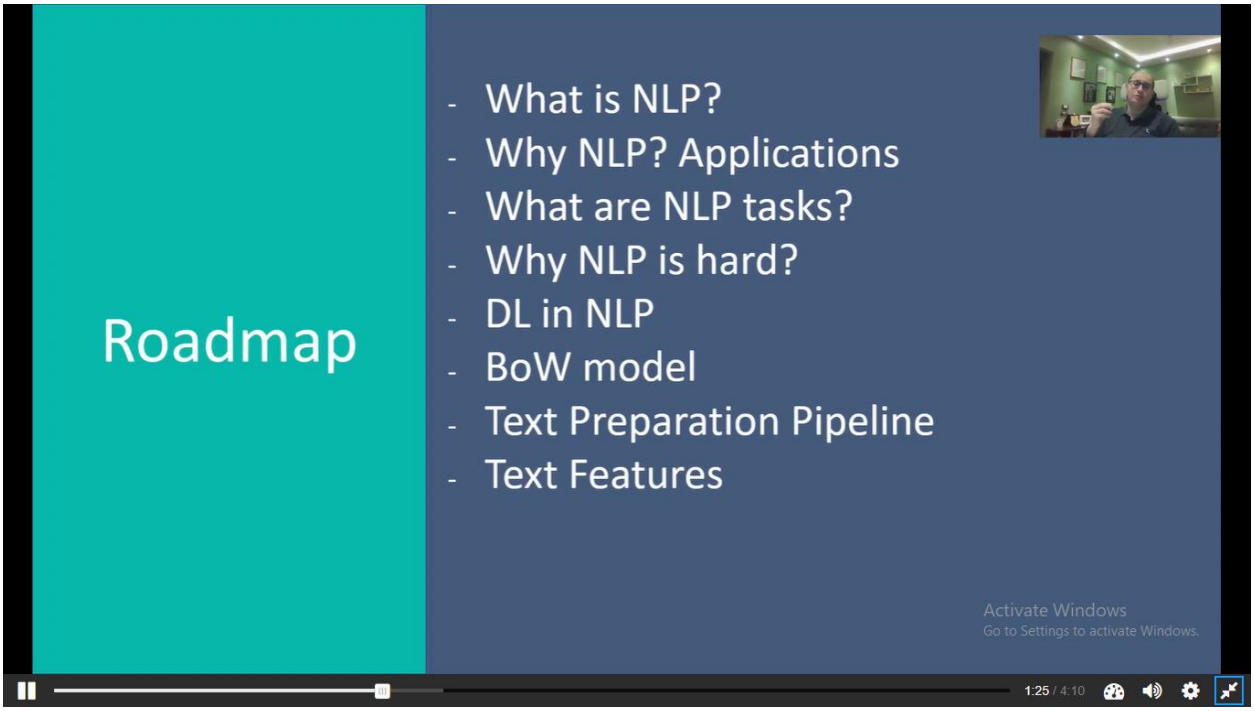
Roadmap
- What is NLP?
- Why NLP? Applications
- What are NLP tasks?
- Why NLP is hard?
- DL in NLP
- BoW model
- Text Preparation Pipeline
- Text Features
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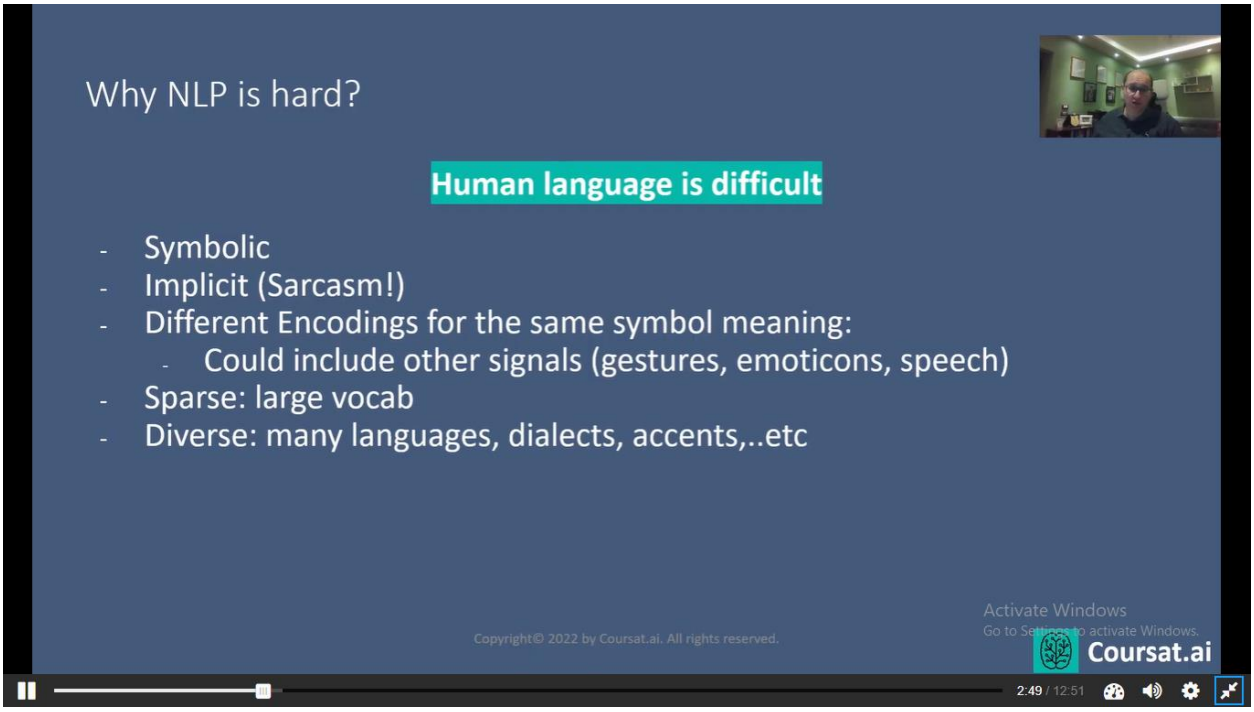
Why NLP is hard?
Human language is difficult
- Symbolic
- Implicit (Sarcasm!)
- Different Encodings for the same symbol meaning:
  - Could include other signals (gestures, emoticons, speech)
- Sparse: large vocab
- Diverse: many languages, dialects, accents,..etc
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What is NLP?

Natural Language => Means/**Media** of communication between Humans:

-Verbal → Speech

-Textual

-Sometimes: visual (Visual QA, **OCR**)

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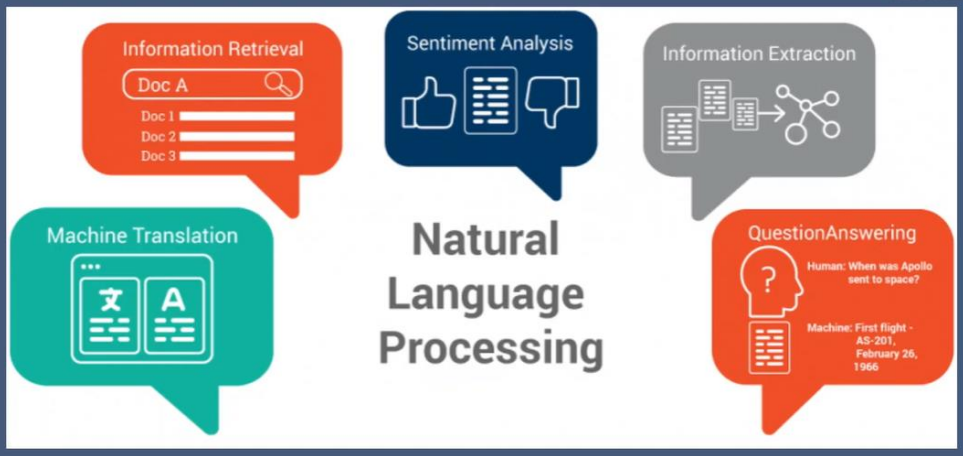
Why NLP?



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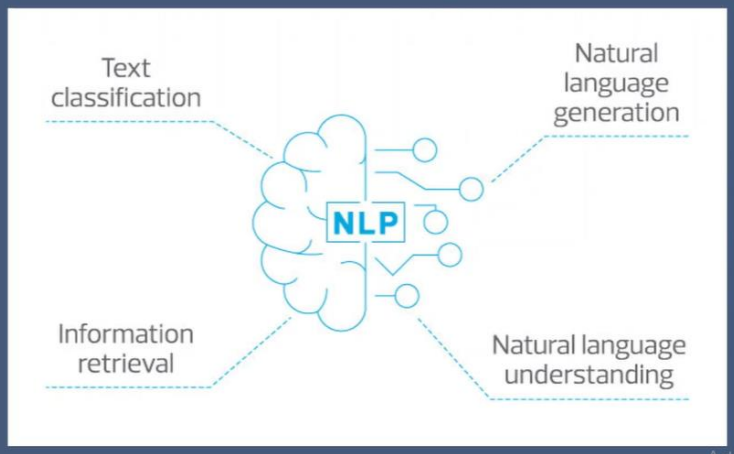
What are NLP tasks?

Sentiment classification
Toxic comments

Information extraction:
Docs parsing → structured DB
Recommender systems



QA
Chatbot
MT
Autocompletion
Summarization

Intent classification

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Encoder-Decoder to close the semantic gap in ML

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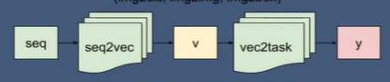
Encoder-Decoder pattern in NLP

(a) Encoder- Decoder Pattern in Deep Learning to close the semantic gap = x2y
The Encoder produces an "Embedding" vector "v" that encoder x → x2v.
The Decoder produces the task y based on the embedding → v2y.

(b) Spatial image case
The Encoder = ConvNet.
Embedding v is a spatial feature map that encodes spatial features in a grid representing regions of input image.
The Decoder depends on the task y (img2cls, img2img, img2box)



(c) Sequential language case
The Encoder = Sequence model (RNN, Transformer,..etc)
Embedding v is a summarization of the sequence.
The Decoder depends on the task y (seq2cls, seq2seq)

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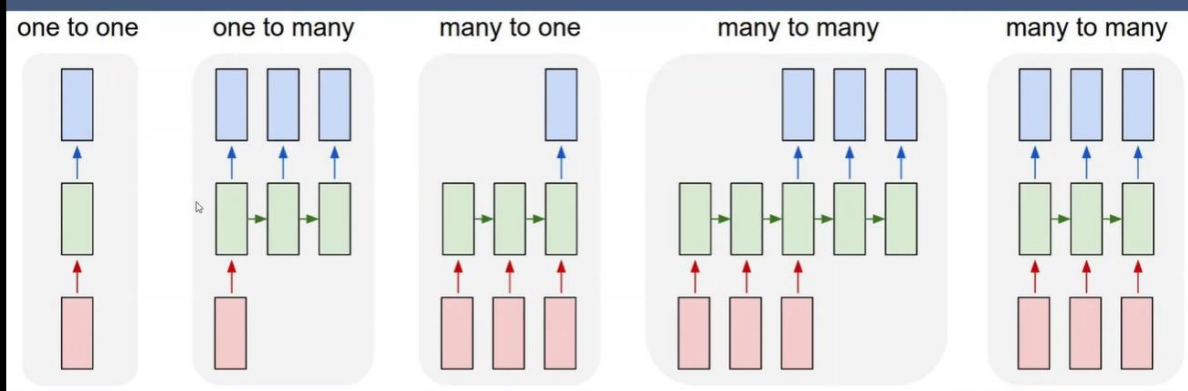
Encoder-Decoder pattern



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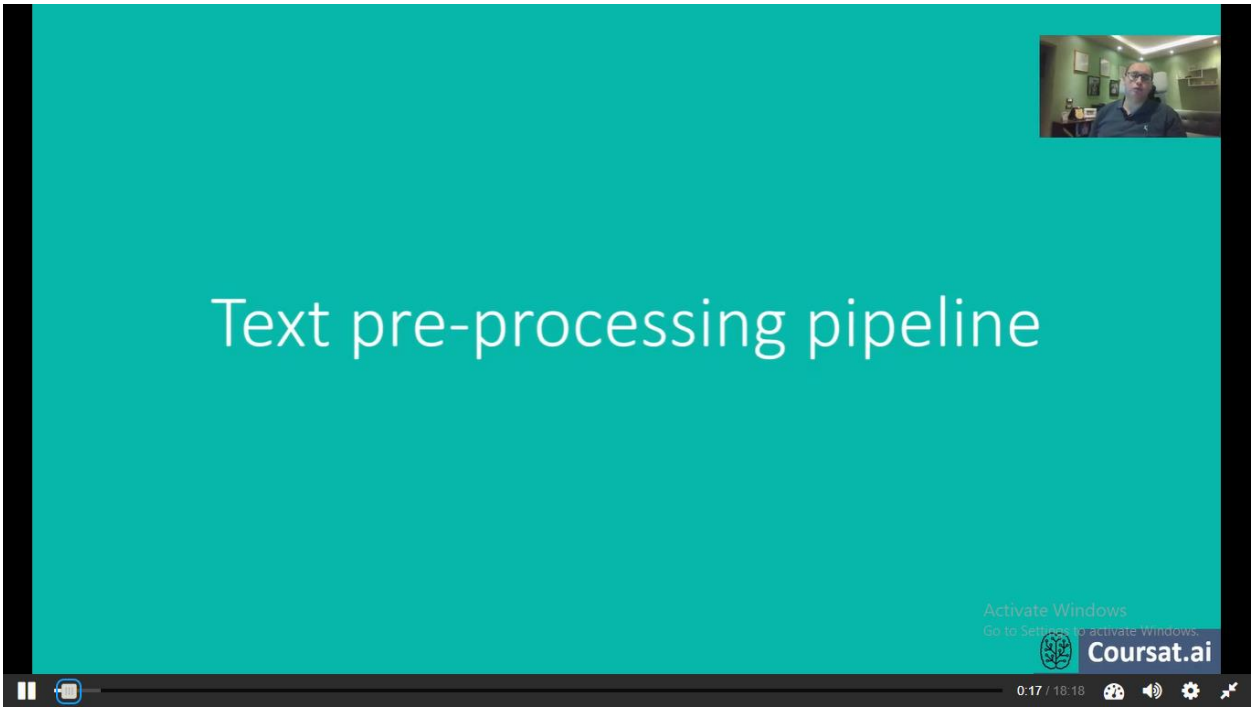
Text pre-processing pipeline
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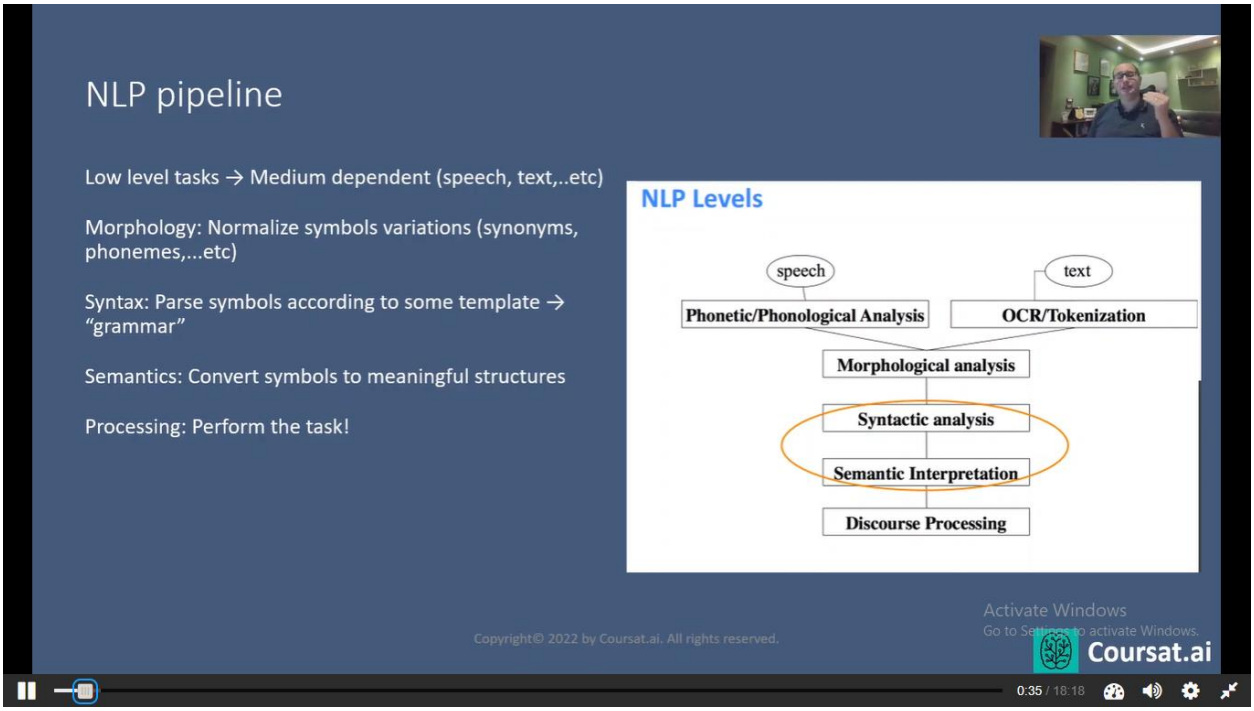
NLP pipeline
Low level tasks → Medium dependent (speech, text,..etc)
Morphology: Normalize symbols variations (synonyms, phonemes,...etc)
Syntax: Parse symbols according to some template → "grammar"
Semantics: Convert symbols to meaningful structures
Processing: Perform the task!
NLP Levels
speech
text
Phonetic/Phonological Analysis
OCR/Tokenization
Morphological analysis
Syntactic analysis
Semantic Interpretation
Discourse Processing
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Two phases

A - Text preprocessing

Text-in→ Text-out

1- **Data sequencing (splitting/tokenization):** each sentence --> sequence (list) of words

2- **Data cleaning:** This step varies from task to task. For some tasks it's better to remove special characters and punctuations, for other they are critical (emotiocons). Good for performance. Sometimes bad! Ex: CAPITALIZED words in sentiment.

3- **Text normalization:** in general text morphology is a big issue in NLP. Upper and lower cases, stemming and lemmatization, ...etc. Again it's task dependent.

4- **Padding (model dependent):** Dense and CNN. RNN can skip this step.

B- Text preparation

Text-in→ Numbers-out

5- **Binarization/vectorization/digitization:** transform words into numbers according to a vocab index.

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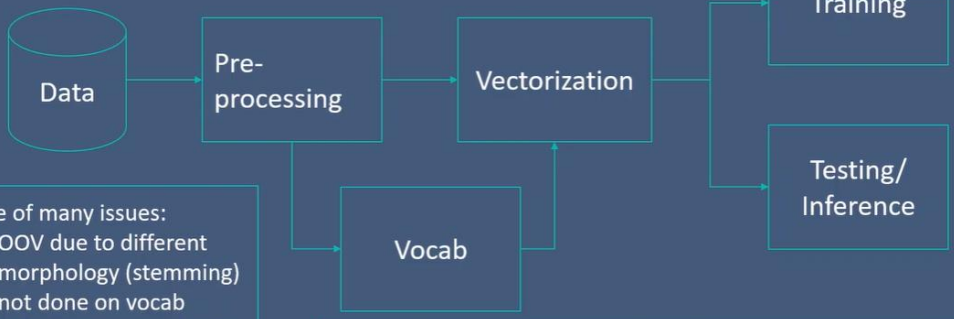
Same pre-processing in training and inference



Source of many issues:
- OOV due to different morphology (stemming) not done on vocab
- Extra large vocab due to non-normalization or non-clean text
- OOV test

As a rule of thumb, perform the same preprocessing on the vocab (or its training corpus), as that on the input text

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Sequencing and Tokenization

- Simple tokenization → split by space
  - Punctuations, Prefixes, Suffixes
  - Missing spaces by mistake
  - Unnecessarily increase vocab size: Hello, Hello!, Hello,...etc
- Handle with regex
- Use special tokenizer
  - NLTK
  - Keras
  - spaCy

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Normalization

- Case-normalization
  - Less vocab → Hello, hello, HELLO → hello
  - Hence, reduce OOV
  - Could remove some meaning → WHAT!
- Remove stop words
  - Again reduces vocab
  - But could remove some meaning → with context and co-reference → No need with sequence models and DL

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PoS Tags

In NLP, Part-of-Speech refers to the different classes a word can belong to: noun, verb, adjective, ...etc. The different tags/classes are called tagset, and there's no common standard. They usually encode grammar + tense.

The task of PoS tagging resembles the task of semantic segmentation in CV; assign a class for every word.

```
s = 'He is going on a journey fishing on ships'
tagged = nltk.pos_tag(s.split())
tagged
```

```
[('He', 'PRP'),
 ('is', 'VBZ'),
 ('going', 'VBG'),
 ('on', 'IN'),
 ('a', 'DT'),
 ('journey', 'NN'),
 ('fishing', 'NN'),
 ('on', 'IN'),
 ('ships', 'NNS')]
```

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Extra clean-up

We might encounter non ascii codes. In this case we need to decode Unicode characters into a normalized form, such as UTF8.

```
import unicodedata
text = 'some text'
unicodedata.normalize('NFKD', text).encode('ascii', 'ignore').decode('utf-8', 'ignore')
```

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Extra clean-up

We might encounter non ascii codes. In this case we need to decode Unicode characters into a normalized form, such as UTF8.

Remove special characters

```
import re
re1 = re.compile(r'  +')
import html

def fixup(x):
    x = x.replace('#39;', "'").replace('amp;', '&').replace('#146;', "'").replace(
        'nbsp;', ' ').replace('#36;', '$').replace('\\n', "\n").replace('quot;', "'").replace(
        '<br />', "\n").replace('\\"', '"').replace('<unk>','u_n').replace(' @.@ ','.').replace(
        ' @-@ ','-').replace('\\', ' \\ ')
    return re1.sub(' ', html.unescape(x))
```

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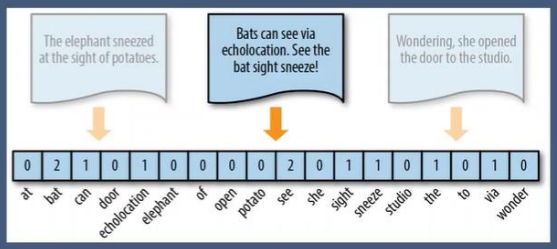
Vectorization

- Build vocab
- Register the index of the word from the vocab
- Not in vocab? → store as UNK
- Need to pad? → encode as PAD
- All special tokens must NOT be used for other symbols
- Example: if PAD=0, don't use 0 as a word index (known mistake)
- Manual or Tokenizer
  - Most tokenizers do both: splitting + vectorization



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Binary features

	1 This	2 movie	3 is	4 very	5 scary	6 and	7 long	8 not	9 slow	10 spooky	11 good	Length of the review(in words)
Review 1	1	1	1	1	1	1	1	0	0	0	0	7
Review 2	1	1	2	0	0	1	1	0	1	0	0	8
Review 3	1	1	1	0	0	0	1	0	0	1	1	6

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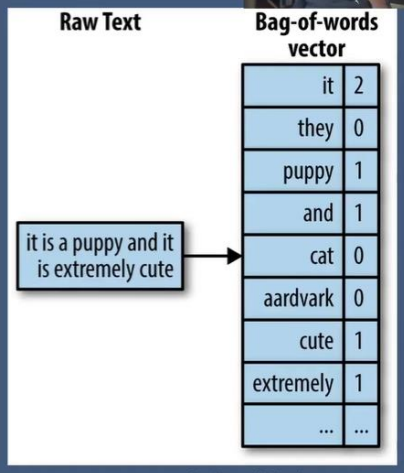
How to encode text data?

First build a vocabulary for your data!

Many ways to encode a review

- BoW: Each sentence will have |V| numbers
  - **Binary:** Put 1/0 if the word is present/absent in a review
  - **Count:** Put the number of times the word is mentioned in a review
  - **Freq:** Normalized counts by total words counts
  - **TF-IDF:** Normalize by the total mentions in all reviews (frequent words are not important) = TF(in rev) x IDF (in ALL revs)
- Sequence (Advanced)



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TF-IDF

$$w_{i,j} = tf_{i,j} \times \log\left(\frac{N}{df_i}\right)$$

$tf_{i,j}$ = number of occurrences of i in j
df_i = number of documents containing i
N = total number of documents

Log → df = 0, log(inf) = 1

N = normalization factor

Bag-of-words (TF)

loved	movie	great	awful
1	1	3	0
1	1	0	1
0	1	0	0
x			

Weights (IDF)

loved	movie	great	awful
log(1/.01)	log(1/.33)	log(1/.01)	0
0	log(1/.33)	0	log(1/.01)
0	log(1/.33)	0	
x			

Term-Document Matrix (TF-IDF Weighted)

loved	movie	great	awful
4.61	1.11	13.82	0
0	1.11	0	4.61
0	1.11	0	0

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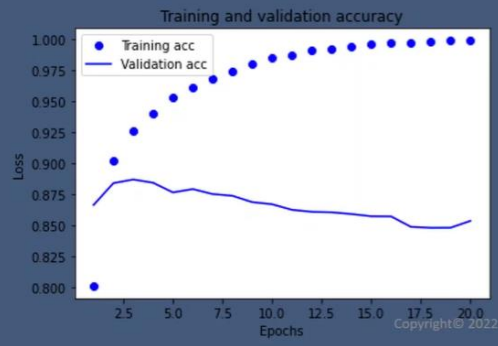
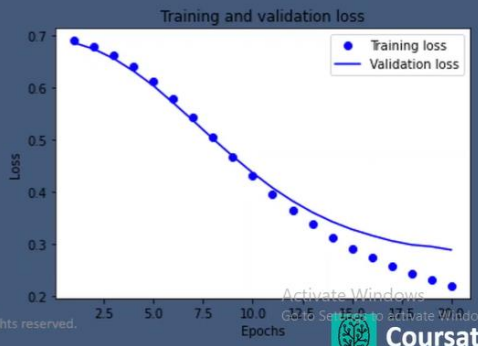
Normalization: Count vs. Freq

As you can see above, the counts are not normalized values, which is not good for the NN (neurons activations prefer normalized values, however, for Dense and Relu it makes no big difference).

However, making just unnormalized counts is the same as binary BoW, since high frequency words will dominate the vector, specially that it's very sparse, causing quick overfitting. In other words, important low freq words are discarded. This can be treated with TF-IDF, or at least by normalizing, so the network might understand that low frequencies have special importance.





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Pros and Cons of BoW

All words are normalized, regardless of their index in the vocab	Sparsity = inefficiency = A lot of useless features = confusion of the model = unnecessary big model = more overfitting
Fast inference → Dense layers	Need to pad to max (again inefficiency)
	No context = no sequence = hard to model co-reference, sarcasm, negation...etc: Example Such BoW model has no clue to differentiate the following cases: • This movie is good --> + • This movie is bad --> - • This movie is not good --> - • This movie is not bad --> + Simple because it cannot link the context of the words "good" and "bad" to the negatition or affirmation context "not"

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What we need?

- A model that encode sequence → sequence models
- But if we use the sent actual word + pad, we end up with features=word index! High and low values + not reflecting the word meaning + no similarity encoding
- So we need:
- A mapping of symbolic words into a space where similarity is encoded → Word Embeddings → Word Vectors

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